



Yakeen Plus (2025)

PRACTICE TEST- 11

DURATION :200 Minutes

DATE : 27/11/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (4)
2. (1)
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CHEMISTRY

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BOTANY

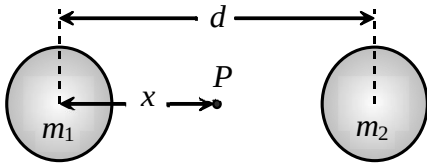
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ZOOLOGY

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200. (3)

SECTION-(I) PHYSICS

1. (4)



Force will be zero at the point of zero intensity

$$x = \frac{\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}} d = \frac{\sqrt{81M}}{\sqrt{81M} + \sqrt{M}} D = \frac{9}{10} D$$

[New NCERT Class 11th Page No. 185]

2. (1)

$$F \propto xm \times (1-x)m = xm^2(1-x)$$

For maximum force $\frac{dF}{dx} = 0$

$$\Rightarrow \frac{dF}{dx} = m^2 - 2xm^2 = 0 \Rightarrow x = 1/2$$

[New NCERT Class 11th Page No. 185]

3. (2)

The value of g at the height h from the surface of earth

$$g' = g \left(1 - \frac{2h}{R} \right)$$

The value of g at depth x below the surface of earth

$$g' = g \left(1 - \frac{x}{R} \right)$$

These two are given equal,

$$\therefore \left(1 - \frac{2h}{R} \right) = \left(1 - \frac{x}{R} \right)$$

On solving, we get

$$x = 2h$$

[New NCERT Class 11th Page No. 190]

4. (3)

$$\Delta U = \frac{mgh}{1 + h/R}$$

Substituting $R = 5h$ we get

$$\Delta U = \frac{mgh}{1 + 1/5} = \frac{5}{6} mgh$$

[New NCERT Class 11th Page No. 191]

5. (4)

$$\text{Gravitational potential} = \int_x^\infty E dx = \int_x^\infty \frac{K}{x^3} dx$$

$$= K \left(\frac{x^{-3+1}}{-3+1} \right)_x^\infty = \left[\frac{-K}{2x^2} \right]_x^\infty = \frac{K}{2x^2}$$

[New NCERT Class 11th Page No. 191]

6. (4)

$$v_0 = \sqrt{\frac{GM}{r}}$$

[New NCERT Class 11th Page No. 193]

7. (1)

Escape velocity is same for all angles of projection.

[New NCERT Class 11th Page No. 190]

8. (3)

$$l = \frac{FL}{AY}$$

$$\Rightarrow l \propto \frac{L}{d^2}$$

$$\Rightarrow \frac{l_1}{l_2} = \frac{L_1}{L_2} \times \left(\frac{d_2}{d_1} \right)^2$$

$$= \frac{1}{2} \times \left(\frac{1}{2} \right)^2 = \frac{1}{8}$$

[New NCERT Class 11th Page No. 170]

9. (4)

$$K = \frac{\Delta P}{\Delta V/V} = \frac{hpg}{\Delta V/V} = \frac{200 \times 10^3 \times 10}{\frac{0.1}{100}} = 2 \times 10^9$$

[New NCERT Class 11th Page No. 173]

10. (4)

$$\frac{Y_A}{Y_B} = \frac{\tan \theta_A}{\tan \theta_B} = \frac{\tan 60}{\tan 30} = \frac{\sqrt{3}}{1/\sqrt{3}} = 3$$

$$\Rightarrow Y_A = 3Y_B$$

[New NCERT Class 11th Page No. 169]

11. (4)

$$L = mvr = m \sqrt{\frac{GM}{r}} r = m \sqrt{GM} r$$

$$\therefore L \propto \sqrt{r}$$

[New NCERT Class 11th Page No. 184]

12. (3)

Due to upward acceleration pseudo force will act downwards so value of acceleration due to gravity will increase by ' a '

$$\therefore g' = (g + a)$$

$$P = \rho g' h$$

$$\Rightarrow P = \rho (g + a) h \quad (\text{Substitute } g')$$

[New NCERT Class 11th Page No. 248]

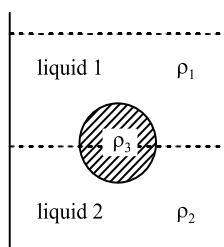
13. (4)

Let each have mass = M and densities d_1 and d_2

$$d_{\text{mix}} = \frac{M_{\text{mix}}}{V_{\text{mix}}} = \frac{M + M}{\left(\frac{M}{d_1} \right) + \left(\frac{M}{d_2} \right)} = \frac{2d_1 d_2}{d_1 + d_2}$$

[New NCERT Class 11th Page No. 246]

14. (1)



The liquid 1 is over liquid 2. Therefore $\rho_1 < \rho_2$.
If ρ_3 had been greater than ρ_2 , it will not be partially inside but anywhere inside liquid 2 if $\rho_3 = \rho_2$ or it would have sunk totally if ρ_3 had been greater than ρ_2 .
 $\therefore \rho_1 < \rho_3 < \rho_2$

[New NCERT Class 11th Page No. 248]

15. (3)

$$\frac{P_A}{P_B} = \frac{P_0 + \rho gh}{P_0 + \rho gh + 2\rho g(2h)} = \frac{10^5 + 10^5}{10^5 + 5 \times 10^5} = \frac{1}{3}$$

[New NCERT Class 11th Page No. 248]

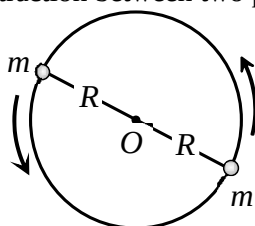
16. (3)

Remains same

[New NCERT Class 11th Page No. 170]

17. (3)

Centripetal force provided by the gravitational force of attraction between two particles.



$$\text{i.e., } \frac{mv^2}{R} = \frac{Gm \times m}{(2R)^2}$$

$$\Rightarrow v = \frac{1}{2} \sqrt{\frac{Gm}{R}}$$

[New NCERT Class 11th Page No. 194]

18. (2)

Pressure in a liquid is divided equally so we can say pressure at both the pistons should be same

$$\Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \left\{ \begin{array}{l} \text{Where,} \\ F_1 = F \\ A_1 = 10 \text{ cm}^2 \\ A_2 = 10 \text{ m}^2 = 10 \times 10^4 \text{ cm}^2 \\ F_2 = 8000 \text{ N} \end{array} \right.$$

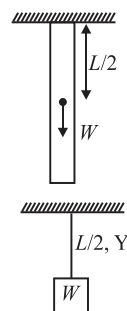
Substituting values,

$$\frac{F}{10} = \frac{8000}{10 \times 10^4}$$

$$\Rightarrow F = 0.8 \text{ N} \quad \left\{ \begin{array}{l} \text{Take} \\ g = 10 \text{ m/s}^2 \end{array} \right.$$

[New NCERT Class 11th Page No. 251]

19. (2)



Let W be the total weight acting downwards
Let the centre of mass which is at the distance of $\frac{L}{2}$ from the top.

So, it can be assumed that a weight W is hung by a massless wire of length $\frac{L}{2}$, Young's modulus Y ,

$$\rho = \text{density of wire, } W = Mg = \rho \times \pi r^2 L \times g$$

$$\text{Using } \Delta L = \frac{FL}{2AY} = \frac{\rho \pi r^2 L g}{\pi r^2} \times \frac{L}{2} \times \frac{1}{Y} = \frac{\rho g L^2}{2Y}$$

[NCERT Class 11th Page No. 170]

20. (1)

$$Q_1 > Q_2 > Q_3 \text{ and } \Delta U_1 = \Delta U_2 = \Delta U_3$$

[New NCERT Class 11th Page No. 307]

21. (4)

By kepler 2nd law

$$\frac{dA}{dt} = \text{constant}$$

$$\frac{A_1}{t_1} = \frac{A_2}{t_2}$$

$$\frac{2A}{t_1} = \frac{A}{t_2}$$

$$t_1 = 2t_2$$

[NCERT Class 11th Page No. 128]

22. (3)

$$K = \frac{GmM}{2R}$$

$$U = -\frac{GmM}{R} = -2K$$

$$E = -\frac{GmM}{2R} = -K$$

$$\frac{K + U + E}{K - U} = \frac{K - 2K - K}{K + 2K} = \frac{-2K}{3K} = \frac{-2}{3}$$

[NCERT Class 11th Page No. 135]

23. (2)

$$\frac{\Delta A}{\Delta t} = \text{constant}$$

$$\Rightarrow \frac{A_1}{3} = \frac{A_2}{6} = \frac{A_3}{2}$$

$$\Rightarrow 2A_1 = A_2 = 3A_3$$

[NCERT Class 11th Page No. 132]

24. (2)
Let a small element dy at a distance y from the mass m be taken, then force due to this element
Gravitational force on the mass due to rod,

$$F = \int_x^{x+L} \frac{GMdy}{Ly^2}$$

$$= \frac{GMm}{L} \left[\frac{L}{x} - \frac{L}{x+L} \right] = \frac{GMm}{x(x+L)}$$

[NCERT Class 11th Page No. 128]

25. (1)

$$I = \frac{GM}{R_e^3} r = \frac{\pi G \rho R_e}{3}$$

[NCERT Class 11th Page No. 134]

26. (1)

$$P + rgh + \frac{1}{2} \rho V^2 = \text{constant}$$

27. (2)

Equivalent spring constant of a wire is given by

$$K = \frac{YA}{l}$$

$$K_{eq} = K_1 + K_2$$

$$\text{or } \frac{Y(2A)}{l} = \frac{Y_1 A}{l} + \frac{Y_2 A}{l}$$

$$\text{or } Y = \frac{Y_1 + Y_2}{2}$$

28. (3)
If the liquid is incompressible then mass of liquid entering through left end, should be equal to mass of liquid coming out from the right end.
 $\therefore M = m_1 + m_2 \Rightarrow Av_1 = Av_2 + 1.5Av$
 $\Rightarrow A \times 3 = A \times 1.5 + 1.5Av \Rightarrow v = 1 \text{ m/s}$
[New NCERT Class 11th Pg No. 187]

29. (4)

$$\tan \theta = \frac{a}{g} = \frac{h}{L}$$

$$\frac{a}{g} = \frac{h}{L}$$

$$h = \frac{aL}{g}$$

[New NCERT Class 11th Pg No. 187]

30. (2)

$$r = \frac{r_1 r_2}{r_1 - r_2} = \frac{5 \times 4}{5 - 4} = 20 \text{ cm}$$

[New NCERT Class 11th Pg No. 195]

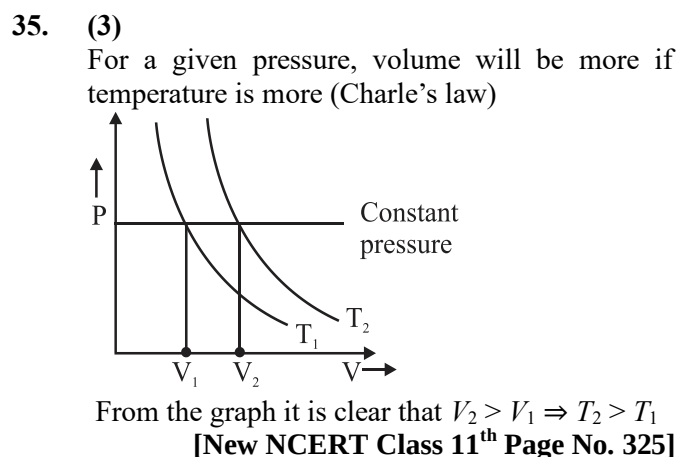
31. (2)
 $AV = \text{constant}$
 $A_1 v_1 = A_2 v_2$
 $\frac{v_1}{v_2} = \frac{A_2}{A_1}$

[New NCERT Class 11th Pg No. 188]

32. (1)
In isochoric processes volume remains constant.
[New NCERT Class 11th Page No. 310]

33. (3)
Resultant amplitude $= \sqrt{3^2 + 4^2} = 5$
[New NCERT Class 11th Page No. 376]

34. (2)
 $\Delta Q = \Delta U + \Delta W$; $\Delta Q = 200 \text{ J}$ and $\Delta W = -100 \text{ J}$
 $\Rightarrow \Delta U = \Delta Q - \Delta W = 200 - (-100) = 300 \text{ J}$
[New NCERT Class 11th Page No. 307]



36. (4)
Clockwise $\rightarrow$ +ve work
Anticlockwise $\rightarrow$ -ve work
 $\Rightarrow$ Work done = 0
[New NCERT Class 11th Page No. 315]

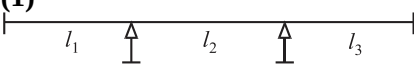
37. (3)
Cyclic process is anticlockwise then
Work done = -(Area of P-V graph)
 $W = -\pi R_1 R_2$
 $= -\pi \left(\frac{3P_0 - P_0}{2} \right) \times \left(\frac{3V_0 - V_0}{2} \right)$
 $= -\frac{22}{7} P_0 V_0$
[New NCERT Class 11th Page No. 311]

38. (3)
Work done = Area under P-V graph
 $= -2P_0 V_0$
[New NCERT Class 11th Page No. 310]

39. (2)
More is value of γ more will be the slope hence, first is O_2 and 2nd is He.
[New NCERT Class 11th Page No. 328]

40. (2)
More is the area under P-V curve more will be the work done in that process.
 $W_1 < W_2 < W_3$
[New NCERT Class 11th Page No. 310]
41. (3)
$$n = \frac{1}{2\pi} \sqrt{\frac{K_{\text{effective}}}{m}} = \frac{1}{2\pi} \sqrt{\frac{(K+2K)}{m}}$$
$$= \frac{1}{2\pi} \sqrt{\frac{3K}{m}}$$

[New NCERT Class 11th Page No. 352]
42. (2)
The collision of molecules of ideal gas is elastic collision.
[New NCERT Class 11th Page No. 323]
43. (3)
[T = constant]
$$\Rightarrow \frac{(v_{\text{rms}})_{\text{O}_2}}{(v_{\text{rms}})_{\text{H}_2}} \sqrt{\frac{M_{\text{H}_2}}{M_{\text{O}_2}}} \Rightarrow \frac{400}{(v_{\text{rms}})_{\text{H}_2}} = \sqrt{\frac{2}{32}} = \frac{1}{4}$$
$$\Rightarrow (v_{\text{rms}})_{\text{H}_2} = 1600 \text{ m/s}$$

[New NCERT Class 11th Page No. 328]
44. (4)
$$v = \frac{\omega}{k}$$
$$v = \frac{4\pi}{\pi/2}$$
$$v = 8 \text{ m/s}$$
45. (1)
$$\frac{(n+1)}{n} = \frac{420}{315}$$
$$\frac{n+1}{n} = \frac{84}{63}$$
$$63n + 63 = 84n$$
$$63 = 21n$$
$$n = 3$$
$$\frac{1}{3} \text{ of } 315 \text{ is fundamental.}$$
$$f = 105$$
46. (1)

$$l = l_1 + l_2 + l_3$$
$$\frac{1}{2n} \sqrt{\frac{T}{\mu}} = \frac{1}{2n_1} \sqrt{\frac{T}{\mu}} + \frac{1}{2n_2} \sqrt{\frac{T}{\mu}} + \frac{1}{2n_3} \sqrt{\frac{T}{\mu}}$$
$$\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$$
$$n_1 = \frac{1}{nl_1} \sqrt{\frac{T}{\mu}} \Rightarrow l_1 = \frac{1}{2n_1} \sqrt{\frac{T}{\mu}}$$
$$n_2 = \frac{1}{2l_2} \sqrt{\frac{T}{\mu}} \Rightarrow l_2 = \frac{1}{2n_2} \sqrt{\frac{T}{\mu}}$$

$$n_3 = \frac{1}{2l_3} \sqrt{\frac{T}{\mu}} \Rightarrow l_3 = \frac{1}{2n_3} \sqrt{\frac{T}{\mu}}$$

$$n_1 = \frac{1}{nl_1} \sqrt{\frac{T}{\mu}} \Rightarrow l_1 = \frac{1}{2n_1} \sqrt{\frac{T}{\mu}}$$

$$n_2 = \frac{1}{2l_2} \sqrt{\frac{T}{\mu}} \Rightarrow l_2 = \frac{1}{2n_2} \sqrt{\frac{T}{\mu}}$$

$$n_3 = \frac{1}{2l_3} \sqrt{\frac{T}{\mu}} \Rightarrow l_3 = \frac{1}{2n_3} \sqrt{\frac{T}{\mu}}$$

47. (2)
$$y = 3 \sin \frac{\pi}{2} (50t - x)$$
$$v_{\text{max}} = Aw$$
$$\text{and}$$
$$v = f\lambda$$
$$\frac{v_{\text{max}}}{v} = \frac{Aw}{f\lambda}$$
$$\frac{v_{\text{max}}}{v} = \frac{A2\pi f}{f\lambda} = \frac{2\pi A}{\lambda}$$
$$\therefore k = \frac{2\pi}{\lambda}$$
$$\frac{v_{\text{max}}}{v} = kA$$
$$= \frac{\pi}{2} \times 3 = \frac{3\pi}{2}$$
48. (2)
$$v = 2(v)[L_2 - L_1]$$
$$= 2 \times 320 [73 - 20] \times 10^{-2}$$
$$= 339.2 \text{ m s}^{-1}$$
$$= 339 \text{ m/s}$$
49. (2)
Two successive frequencies of closed pipe
$$\frac{nv}{4l} = 220 \quad \dots(i)$$
$$\frac{(n+2)v}{4l} = 260 \quad \dots(ii)$$

Dividing (ii) by (i), we get
$$\frac{n+2}{n} = \frac{260}{220} = \frac{13}{11}$$
$$11n + 22 = 13n$$
$$n = 11$$

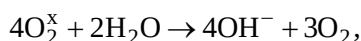
So, $11 \frac{v}{4l} = 220$
$$\frac{v}{4l} = 20$$

So fundamental frequency is 20 Hz.
50. (3)
If $y = f(ax \pm bt)$
then $v = \left| \frac{b}{a} \right|$
$$y = \frac{9}{3 + (x - 10t)^2}$$
$$v = 10 \text{ m/s}$$

SECTION-(II) CHEMISTRY

51. (4)

- The given reaction:



- The total charge of LHS is $4x$, as water molecule is electrically neutral, so the charge of water molecule is zero.
- The total charge of RHS is -4 , as here the charge of oxygen is zero.
- Hence, $4x = -4 \Rightarrow x = -1$.
- Hence $x = -1$ and the species is superoxide as the formula of superoxide is O_2^{-1} .
- The correct option is [4].

[New NCERT Class 11th Page No. 237]

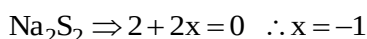
52. (1)

In the given reaction total 3 electrons are involved in both oxidation half and reduction half of the reaction.

The equivalent weight is equal to its formula weight or molecular weight divided by the n -factor.

Thus the equivalent weight of ferrous oxalate is given by $M/3$.

53. (3)

[New NCERT Class 11th Page No. 239]

54. (1)

In this reaction H_2S works a reducing agent while H_2O_2 serves as an oxidizing agent.

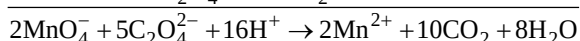
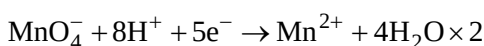
[New NCERT Class 11th Page No. 237]

55. (1)

The removal of an electropositive element from any chemical species is referred to as oxidation. Also, oxidation is a chemical reaction in which hydrogen is removed from any chemical species and an electronegative element is added to any chemical species.

[New NCERT Class 11th Page No. 237]

56. (3)

[New NCERT Class 11th Page No. 247]

57. (1)

The correct match is as follows

Combination reactions	$\text{C(s)} + \text{O}_2\text{(g)} \xrightarrow{\Delta} \text{CO}_2\text{(g)}$
Decomposition reactions	$2\text{NaH(s)} \xrightarrow{\Delta} 2\text{Na(s)} + \text{H}_2\text{(g)}$
Non-metal displacement reactions	$2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
Disproportionation reactions	$\text{P}_4\text{(s)} + 3\text{OH}^-\text{(aq)} \rightarrow \text{PH}_3\text{(g)} + 3\text{H}_2\text{PO}_2^-\text{(aq)}$

[New NCERT Class 11th Page No. 244]

58. (1)

Purely ionic

[New NCERT Class 12th Page No. 132]

59. (4)

Molar conductance of $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ is zero as it does not have any ionisable chloride ions.

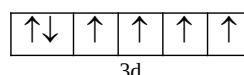
[New NCERT Class 12th Page No. 119]

60. (1)

Electronic configuration of CO^{3+} is $[\text{Ar}]\text{3d}^6$

As F^- is a weak field ligand, no pairing occurs.

No. of unpaired electrons in $[\text{COF}_6]^{3-} = 4$

[New NCERT Class 12th Page No. 129]

61. (2)

$$E_3 = \frac{E_1 n_1 + E_2 n_2}{n_3}$$

[New NCERT Class 12th Page No. 35]

62. (4)

Hydrogen and oxygen are liberated in the ratio 2:1.

[New NCERT Class 12th Page No. 53]

63. (3)

It is not applicable for weak electrolyte.

[New NCERT Class 12th Page No. 47]

64. (3)

$$E^\circ = (2 \times 0.337) - (1 \times 0.153) = 0.521\text{V}$$

[New NCERT Class 12th Page No. 35]

65. (3)
Potential gradient can be defined as the voltage of cell where distance between the electrodes is unity
[New NCERT Class 12th Page No. 33]

66. (1)
Molar conductivity increases with decrease in concentration.
On decreasing the concentration, the total volume (V), of solution containing one mole of electrolyte increases.
[New NCERT Class 12th Page No. 47]

67. (3)
Crystal Field Stabilization Energy (CFSE) is the additional stabilization gained by the splitting of the d-orbitals. The tetrahedral CFSE is expressed as Δ_t and octahedral CFSE is expressed as Δ_o . In crystal field theory, it can be shown that the amount of splitting in a tetrahedral field is much smaller than in an octahedral field. In general,
$$\Delta_t = \frac{4}{9} \Delta_o.$$

[New NCERT Class 12th Page No. 133]

68. (3)
EDTA is a hexadentate ligand.
[New NCERT Class 12th Page No. 123]

69. (1)
When dissolved in water, a double salt dissociates into the two individual salts that make it up. $K_4[Fe(CN)_6]$ does not dissociate into Fe^{2+} and CN^- ions
[New NCERT Class 12th Page No. 120]

70. (4)
Valency = oxidation number
 $X + 0 = +2$. Hence $X = +2$
There are 4 monodentate ammonia ligands forming 4 coordinate bonds. Hence, coordination number is 4.
[New NCERT Class 12th Page No. 121]

71. (2)
$$CH_3CH=CH_2 + O_3 \longrightarrow CH_3-CH \begin{array}{c} \diagup O \diagdown \\ | \quad | \\ O-O \end{array} CH_2$$

$$\downarrow Zn/H_2O$$

$$CH_3CHO + HCHO$$

Propene on ozonolysis gives formaldehyde and acetaldehyde.
[New NCERT Class 11th Page No. 313]

72. (2)
The reaction is as follows
$$CH_3-CH_2-\underset{\substack{| \\ OH}}{CH}-CH_3 \xrightarrow{H_2SO_4} CH_3-CH_2-\underset{\substack{| \\ \oplus OH_2}}{CH}-CH_3$$

$$\xrightarrow{-H_2O} CH_3-\overset{\substack{xH \\ |}}{CH}-\overset{\oplus}{CH}-\underset{\substack{| \\ H_y}}{CH_2} \xrightarrow{-H_x^+} CH_3-CH=CH-CH_3 \quad (\text{Major product})$$

$$\downarrow \Delta$$

$$CH_3-CH_2-CH=CH_2 \quad (\text{Minor product})$$

[New NCERT Class 11th Page No. 310]

73. (3)
• Sodamide is a strong base used in organic synthesis. It is a nucleophilic reagent.
• On heating 2-butyne with $NaNH_2$ in an inert solvent, the sodium derivative of but-1-yne is converted into but-1-yne by the action of dil. HCl.
• The concerned reaction is as follows;
$$CH_3-C \equiv C-CH_3 + NaNH_2 \xrightarrow{\text{Paraffin}} NH_3 + CH_3CH_2C \equiv CNa \xrightarrow{\text{dil. HCl}} CH_3CH_2C \equiv CH$$

1-butyne
• The final product formed is 1-butyne. Hence, correct option is 3.
[New NCERT Class 11th Page No. 317]

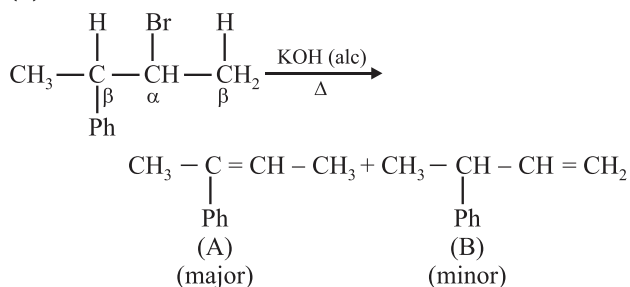
74. (2)
$$\begin{array}{c} CH_3 \\ | \\ CH_2 \\ | \\ CH_2-CH \\ | \quad | \\ CH_2 \quad CH_3 \end{array} \xrightarrow[773 \text{ K, } 10-20 \text{ atm}]{Mo_2O_3} \text{C}_6\text{H}_5-CH_3$$

Toluene
[New NCERT Class 11th Page No. 304]

75. (3)
• The bromination of alkane follows a free radical mechanism which occurs in presence of sunlight.
• Free radicals contain unpaired electrons in its valence shell.
• The bromination of propane (C_3H_8) in sunlight yields two monobromo alkane product and HBr is formed.
• The concerned reaction is as follows;
$$CH_3-CH_2-CH_3 \xrightarrow{Br_2/\text{light}} CH_3-CH_2-\underset{\substack{| \\ Br}}{CH_2} + CH_3-\underset{\substack{| \\ Br}}{CH}-CH_3 + HBr$$

1-Bromo propane
2-Bromo propane
• So the required hydrocarbon X is propane (C_3H_8)
[New NCERT Class 11th Page No. 303]

76. (1)



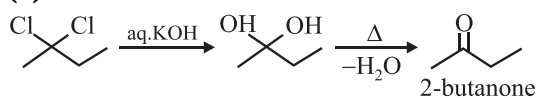
[New NCERT Class 12th Page No. 313]

77. (3)

Tertiary alkyl halides form more stable carbocations, which is why they undergo faster S_N1 reactions.

[New NCERT Class 11th Page No. 302]

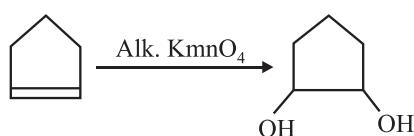
78. (2)



[New NCERT Class 12th Page No. 173]

79. (1)

- Cyclopentene is a cyclic organic compound.
- Alkaline KMnO₄ is a strong oxidising agent which converts the alkenes into diols.
- There is a formation of intermediate on reaction with MnO₄⁻. Further in presence of alkaline medium, OH⁻ attacks and forms the required product.
- The concerned reaction is as follows;



- The product formed is cis 1,2 - cyclopentanediol

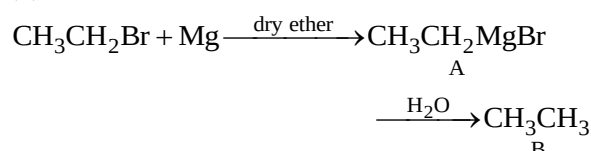
[New NCERT Class 11th Page No. 313]

80. (1)

Boiling point of alkene is lower than alkane boiling point.

[New NCERT Class 11th Page No. 301]

81. (1)



[New NCERT Class 12th Page No. 181]

82. (2)

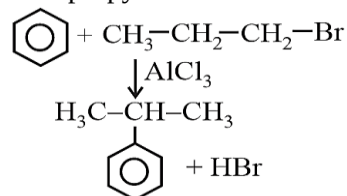
The longest carbon chain is 3 carbon long and 'Cl' substituent is present at 2nd position along with methyl group.

∴ The correct IUPAC name is 2-chloro-2-methylpropane.

[New NCERT Class 12th Page No. 170]

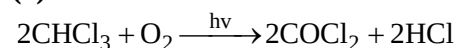
83. (3)

Propyl bromide reacts with AlCl₃ to form stable isopropyl carbocation. This leads to the formation of isopropyl benzene.



[New NCERT Class 12th Page No. 180]

84. (3)



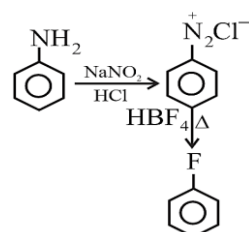
[New NCERT Class 12th Page No. 187]

85. (2)

Bromobenzene cannot be substituted with NaF to form fluorobenzene.

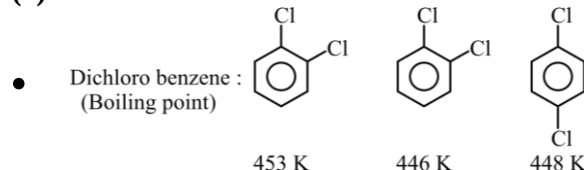
When aniline is diazotised, it produces benzene diazonium chloride.

When this is treated with HBF₄, it produces fluorobenzene.



[New NCERT Class 12th Page No. 166]

86. (1)



- The boiling points of isomeric Haloalkanes decreases with increase in branching.

[New NCERT Class 12th Page No. 169]

87. (1)

(A → III; B → IV; C → I; D → II)

[A → III] KMnO₄ gives syn addition

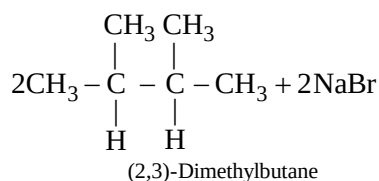
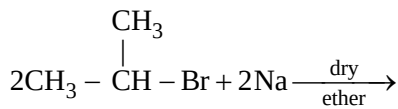
[B → IV] Br₂ gives anti-addition

[C → I] R₂O₂ (peroxide) gives free radical mechanism, which gives anti-Markovnikov's product.

[D → II] Hot acidic KMnO₄ is very strong oxidizing agent.

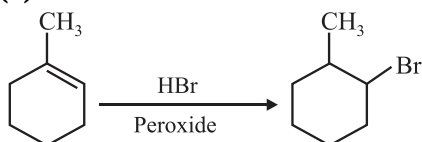
[New NCERT Class 11th Page No. 312]

88. (3)



[New NCERT Class 11th Page No. 301]

89. (2)



[New NCERT Class 11th Page No. 311]

90. (4)

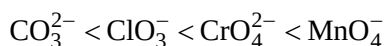
$\text{CH}_3 - \underset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_2\text{Br}$ is known as iso-butyl bromide.

[New NCERT Class 12th Page No. 161]

91. (1)

Species	Oxidation state of central atom
CO_3^{2-}	+4
ClO_3^-	+5
CrO_4^{2-}	+6
MnO_4^-	+7

Correct increasing order of oxidation number of the central atom is



[New NCERT Class 11th Page No. 240]

92. (2)

Let the oxidation state of carbon be x.

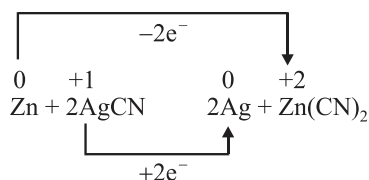
$$2 + x - 4 = 0$$

$$x = 2$$

[New NCERT Class 11th Page No. 241]

93. (4)

The oxidation states show a change only in given reaction.



[New NCERT Class 11th Page No. 237]

94. (3)

The metal having higher reduction potential will get reduced, and oxidize other metals.

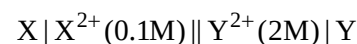
[New NCERT Class 12th Page No. 37]

95. (2)

Electrode potential of both electrodes becomes equal.

[New NCERT Class 12th Page No. 32]

96. (3)



$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$$

$$= 0.34 - (-0.25)$$

$$= 0.59 \text{ V}$$

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{n} \log \left[\frac{\text{X}^{2+}}{\text{Y}^{2+}} \right]$$

$$E_{\text{cell}} = 0.59 - \frac{0.059}{2} \log \frac{0.1}{1}$$

$$= 0.59 + \frac{0.059}{2} = 0.62 \text{ V}$$

[New NCERT Class 12th Page No. 35]

97. (2)

In fused NaCl chloride ions are oxidized at anode and it is called oxidation.

[New NCERT Class 12th Page No. 53]

98. (2)

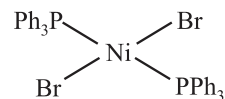
If $\Delta_0 < P$ in octahedral system then, e^- goes from t_{2g} to e_g level without getting paired.

i.e., d^4 system will be: $t_{2g}^3 e_g^1$

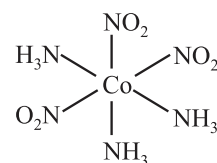
[New NCERT Class 12th Page No. 134]

99. (1)

(a) The structures are:



Meridional - $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$



[New NCERT Class 12th Page No. 125]

100. (1)

cis - $[\text{PtCl}_2(\text{NH}_3)_2]$ is used as a anticancer agent.

[New NCERT Class 12th Page No. 137]

SECTION-(III) BOTANY

101. (4)

Xylem parenchyma cells are living and thin-walled, and their cell walls are made up of cellulose.

[Old NCERT Class 11th Page No. 87]

102. (3)

In isobilateral leaf, the stomata are present on both the surfaces of the epidermis; and the mesophyll is not differentiated into palisade and spongy parenchyma. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells.

[Old NCERT Class 11th Page No. 94]

103. (4)

Palisade parenchyma is present below the adaxial epidermis in dicot leaves.

[Old NCERT Class 11th Page No. 93]

104. (2)

As compared to the dicot root that have fewer xylem bundles, there are usually more than six (polyarch) xylem bundles in the monocot root.

[Old NCERT Class 11th Page No. 91]

105. (3)

A and D are correct.

Each stoma is composed of two bean shaped cells known as guard cells which enclose stomatal pore. The guard cells possess chloroplasts and regulate the opening and closing of stomata.

[Old NCERT Class 11th Page No. 89]

106. (3)

Pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma.

[Old NCERT Class 11th Page No. 93]

107. (1)

Dicot stem	Hypodermis present
Monocot root	Pith is well developed
Leaves	Mesophyll
Dicot root	Pith is small

[Old NCERT Class 11th Page No. 91, 92, 93]

108. (2)

- From the region of maturation, some of the epidermal cells form root hairs.
- Maize and sugarcane have stilt roots.

[Old NCERT Class 11th Page No. 67]

109. (1)

<i>Calotropis</i>	Opposite phyllotaxy
Sunflower	Alternate phyllotaxy
<i>Alstonia</i>	Whorled phyllotaxy
Rachis	Pinnately compound leaf

[Old NCERT Class 11th Page No. 71]

110. (3)



Given figure of root is of fibrous roots of wheat

[Old NCERT Class 11th Page No. 66]

111. (3)

Pneumatophores grow upwards (negative geotropism) and have pores (lenticels) that allow gaseous exchange, ensuring the plant receives sufficient oxygen for respiration in oxygen-deficient swampy environments.

[Old NCERT Class 11th Page No. 67]

112. (3)

- Androecium and gynoecium are reproductive organs of flower.
- When a flower either has only stamens or only carpels it is said to be unisexual.

[Old NCERT Class 11th Page No. 72]

113. (4)

All statements are true.

[Old NCERT Class 11th Page No. 62, 63]

114. (1)

The family Solanaceae is characterized by the following features:

- Leaves having alternate phyllotaxy: Solanaceae plants typically exhibit alternate leaf arrangement.
- Bisexual and actinomorphic flowers: The flowers are perfect (bisexual) and radially symmetrical (actinomorphic).
- Five epipetalous stamens: Stamens are attached to the petals (epipetalous) and usually five in number.
- Fruits are berry or capsule: Solanaceae fruits can either be fleshy berries (e.g., tomato) or dry capsules (e.g., Datura).
- Seeds are endospermic: The seeds contain endosperm, which serves as the food reserve.

[Old NCERT Class 11th Page No. 80]

115. (2)

During glycolysis, the conversion of 2-phosphoglycerate to phosphoenolpyruvate is catalysed by the enzyme enolase. This reaction involves the removal of a water molecule (dehydration),

[New NCERT Class 11th Page No. 156]

116. (4)

Alcoholic fermentation releases CO₂ along with ethanol.

[New NCERT Class 11th Page No. 157]

117. (3)

- Krebs' cycle takes place in the matrix of mitochondria.
- Pyruvate is transported from the cytoplasm into the mitochondria.

[New NCERT Class 11th Page No. 157]

118. (3)

- Glucose is the primary molecule that enters glycolysis directly.
- Fructose enters glycolysis after being converted to intermediates like fructose-6-phosphate or fructose-1-phosphate, which can directly integrate into the glycolytic pathway.

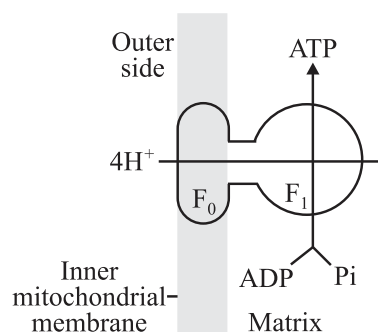
[New NCERT Class 11th Page No. 156]

119. (2)

Cytochrome *c* is a small protein attached to the outer surface of the inner membrane and acts as a mobile carrier for the transfer of electrons between complex III and complex IV.

[New NCERT Class 11th Page No. 160]

120. (3)



Diagrammatic presentation of ATP synthesis in mitochondria.

[New NCERT Class 11th Page No. 161]

121. (1)

During glycolysis, glucose undergoes partial oxidation to form two molecules of pyruvic acid.

[New NCERT Class 11th Page No. 155]

122. (4)

Four ATP molecules are directly synthesised in glycolysis from one glucose molecule.

[New NCERT Class 11th Page No. 156]

123. (3)

The 'bakanae' (foolish seedling) disease of rice seedlings, was caused by a fungal pathogen *Gibberella fujikuroi*. E. Kurosawa (1926) reported the appearance of symptoms of the disease in rice seedlings when they were treated with sterile filtrates of the fungus.

[New NCERT Class 11th Page No. 175]

124. (3)

Lag phase	Initial phase of growth
Exponential phase	Growth is rapid and maximum
Stationary phase	Growth slows down and become steady
Relative growth	Efficiency index

[New NCERT Class 11th Page No. 170]

125. (3)

E. Kurosawa observed symptoms of disease caused by *Gibberella fujikuroi* in rice seedlings.

[New NCERT Class 11th Page No. 175]

126. (4)

Ethylene and ABA promotes senescence and abscission of plant organs especially of leaves and flowers.

[New NCERT Class 11th Page No. 177]

127. (1)

Auxin	Control xylem differentiation
Abscisic acid	Closure of stomata
Gibberellin	Increases size of apple
Ethylene	Fruit ripening

[New NCERT Class 11th Page No. 176, 177]

128. (3)

By inducing dormancy, ABA helps seeds to withstand desiccation and other factors unfavourable for growth.

[New NCERT Class 11th Page No. 178]

129. (3)

The plant growth hormone that is a purine derivative and helps in adventitious shoot formation is cytokinin.

[New NCERT Class 11th Page No. 176]

130. (4)

If the apical bud (terminal bud) is removed from a plant, the plant will exhibit increased lateral branching and activation of axillary buds.

[New NCERT Class 11th Page No. 175]

131. (2)

C₃ plant have a much lower optimum temperature (20° – 25°C). The C₄ plants respond to a higher temperature (30° – 40°C) and show higher rate of photosynthesis.

[New NCERT Class 11th Page No. 145, 146]

132. (2)

In the bundle sheath cells the C₄ acids are broken down to release CO₂ and a 3-carbon molecule. The CO₂ released in the bundle sheath cells enters the C₃ or the Calvin pathway. The bundle sheath cells are rich in an enzyme Ribulose biphosphate carboxylase-oxygenase (RuBisCO).

[New NCERT Class 11th Page No. 146]

133. (2)

Tomato is C₃ plant.

The Calvin cycle, which is responsible for fixing carbon dioxide into sugar during photosynthesis, requires 3 molecules of ATP and 2 molecules of NADPH for every molecule of CO₂ fixed.

For 24 molecules of carbon dioxide fixed, the requirement would be:

ATP: 24 molecules CO₂ → 3 ATP/CO₂ = 72 ATP

NADPH: 24 molecules CO₂ → 2NADPH/CO₂ = 48 NADPH

[New NCERT Class 11th Page No. 146]

134. (2)

CF₀ is embedded in the thylakoid membranes and forms a transmembrane channel that carries out facilitated diffusion of protons across the membrane.

[New NCERT Class 11th Page No. 142]

135. (2)

During ETS within the chloroplast, protons in the stroma decrease in number, while in the lumen there is accumulation of protons.

[New NCERT Class 11th Page No. 141]

136. (2)

The rolling of leaves during water stress is the primary function of bulliform cells.

[Old NCERT Class 11th Page No. 94]

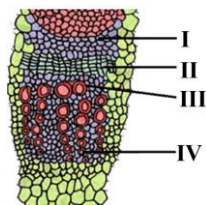
137. (1)

Epidermis is usually single layered.

[Old NCERT Class 11th Page No. 88, 89]

138. (3)

Exarch and endarch conditions are defined based on the position of protoxylem and metaxylem in vascular bundles.



III represents metaxylem and IV represents protoxylem.

[Old NCERT Class 11th Page No. 75]

139. (2)

In the roots of flowering plants, the primary xylem arrangement is exarch, meaning the protoxylem lies towards the periphery, and the metaxylem is oriented towards the center (or pith).

[Old NCERT Class 11th Page No. 74]

140. (4)

Auxin stimulates root formation on stem cutting, particularly the woody ones. It also promotes a lowering in pineapples.

[New NCERT Class 11th Page No. 175]

141. (1)

NAA → naphthalene acetic acid and 2, 4, D (2, 4-dichlorophenoxy acetic acid) are synthetic auxin.

[New NCERT Class 11th Page No. 175]

142. (1)

Ethylene is considered as the plants growth hormone which are involved in growth promoting activities such as cell division cell enlargement etc. They are also involved in various inhibiting.

[New NCERT Class 11th Page No. 177]

143. (3)

I-Epicotyl hook, II-Cotyledon, III-Hypocotyl

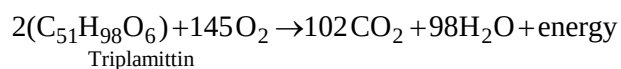
[New NCERT Class 11th Page No. 167]

144. (1)

Glucose is the favoured substrate for respiration.

[New NCERT Class 11th Page No. 162]

145. (3)



$$RQ = \frac{102CO_2}{145O_2} = 0.7$$

[New NCERT Class 11th Page No. 164]

146. (3)

Accessory pigments also absorb light and transfer the energy to chlorophyll a. This helps in capturing a wider range of wavelength of incoming light to be utilised for photosynthesis. Moreover they also protect chlorophyll a from photo-oxidation.

[New NCERT Class 11th Page No. 138]

147. (2)

RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase) is present in both C₃ and C₄ plants.

[New NCERT Class 11th Page No. 143, 147]

148. (2)

Imbricate	Gulmohur
Valvate	<i>Calotropis</i>
Vexillary	Bean
Twisted	Cotton

[New NCERT Class 11th Page No. 64]

149. (3)

Flowers of Malvaceae family are polypetalous (with free petals), and their petals exhibit twisted aestivation.

[Old NCERT Class 11th Page No. 78]

150. (1)

A flower is a modified shoot wherein the shoot apical meristem changes to floral meristem. Internodes do not elongate and the axis gets condensed.

[New NCERT Class 11th Page No. 61]

SECTION-(IV) ZOOLOGY**151. (3)**

- Skin and lungs assist in excretion.
- Nearly 99% of filtrate is reabsorbed through different parts of nephron.
- Dialysing fluid have the same composition as that of plasma except the nitrogenous wastes.

[New NCERT Class 11th Page No. 213, 214]**152. (1)**

The descending limb of loop of Henle is permeable to water but almost impermeable to electrolytes. This concentrates the filtrate as it moves down. The ascending limb is impermeable to water but allows transport of electrolytes actively or passively.

[New NCERT Class 11th Page No. 209]**153. (4)**

The Malpighian corpuscle, PCT and DCT of the nephron are situated in the cortical region of the kidney.

[New NCERT Class 11th Page No. 208]**154. (2)**

- The sweat and sebaceous glands in the skin can eliminate certain substances through their secretions.
- Sweat produced by the sweat glands is a watery fluid containing NaCl, small amounts of urea, lactic acid, etc.
- Sebaceous glands eliminate certain substances like sterols, hydrocarbons and waxes through sebum.

[New NCERT Class 11th Page No. 213]**155. (1)**

Insects	Malpighian tubules
Earthworm	Nephridia
<i>Planaria</i>	Flame cells
Crustaceans	Antennal glands

[New NCERT Class 11th Page No. 206]**156. (3)**

The urine formed is a light yellow coloured watery fluid which is slightly acidic (pH-6.0) and has a characteristic odour.

[New NCERT Class 11th Page No. 213]**157. (3)**

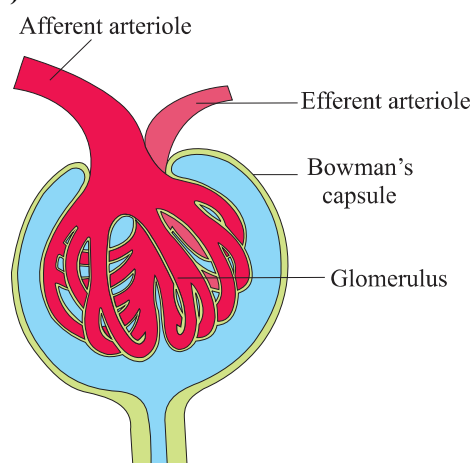
- Presence of glucose in urine is glycosuria and ketone bodies in urine is ketonuria.
- Asthma is a respiratory disorder.

[New NCERT Class 11th Page No. 213]**158. (2)**

- Reptiles, birds, land snails and insects excrete nitrogenous wastes as uric acid.
- Many bony fishes, aquatic amphibians and aquatic insects are ammonotelic.
- Mammals, many terrestrial amphibians and marine fishes mainly excrete urea and are called ureotelic animals.

[New NCERT Class 11th Page No. 205]**159. (4)**

- The tubular epithelial cells in different segments of nephron perform this either by active or passive mechanisms.
- For example, substances like glucose, amino acids, Na⁺, etc., in the filtrate are reabsorbed actively whereas the nitrogenous wastes are absorbed by passive transport.

[New NCERT Class 11th Page No. 209]**160. (1)**

Afferent arteriole is a fine branch of renal artery.

[New NCERT Class 11th Page No. 293]**161. (3)**

Neutrophils	60-65 per cent.
Lymphocytes	20-25 per cent.
Monocytes	6-8 per cent.
Eosinophils	2-3 per cent.
Basophils	0.5-1 per cent.

[New NCERT Class 11th Page No. 279]**162. (4)**

Certain factors released by the tissues at the site of injury can initiate coagulation.

[New NCERT Class 11th Page No. 281]

163. (4)
Open circulatory system is present in the members of phylum arthropoda and mollusca.
[New NCERT Class 11th Page No. 282]

164. (3)
Cardiac output = Stroke volume $\times$ Heart rate
6000 ml = 60 ml $\times$ Heart rate
So, Heart rate = 100 beats per minute
[New NCERT Class 11th Page No. 285]

165. (2)

- Pericardium is the double layer covering of heart.
- Fats are absorbed through lymph in the lacteals present in the intestinal villi.

[New NCERT Class 11th Page No. 286]

166. (1)

Coronary artery disease	Lumen of arteries becomes narrower
Angina pectoris	Insufficient oxygen to heart
Cardiac arrest	Heart stops beating
Heart failure	Heart is not pumping blood effectively enough to meet the needs of the body

[New NCERT Class 11th Page No.286]

167. (1)
Tricuspid valve guards the opening between the right atrium and the right ventricle.
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168. (2)
The Islets of Langerhans are clusters of endocrine cells within the pancreas.
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169. (2)

- Person with AB blood group can receive blood from any other blood group. Blood group AB is called universal recipient.
- Person with B blood group has anti-A antibodies in their plasma.

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170. (1)
Expiratory capacity = Tidal volume + Expiratory reserve volume
EC = TV + ERV
EC = 800 + 1000 = 1800 mL
[New NCERT Class 11th Page no. 185]

171. (3)
The sarcomere is the smallest functional unit of a myofibril and the basic unit of skeletal muscle contraction.
[New NCERT Class 11th Page No. 220]

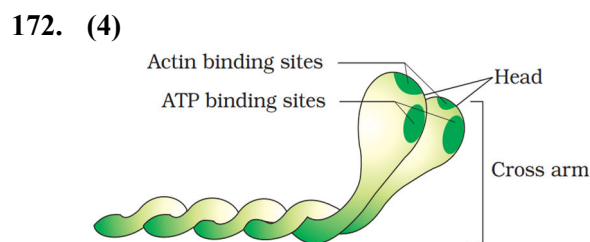


Figure Myosin monomer (Meromyosin)
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173. (3)
The joint between the atlas and axis is a pivot joint.
[New NCERT Class 11th Page No. 227]

174. (3)

- The hypothalamus is the basal part of diencephalon.
- The pituitary gland is in sella tursica.

[New NCERT Class 11th Page No. 240]

175. (3)

- Central nervous system consists of the brain and spinal cord.
- CNS is responsible for processing information, not the PNS.

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176. (1)
Aldosterone is the main mineralocorticoids responsible for reabsorption of Na⁺ and water from the renal distal tubule.
[New NCERT Class 11th Page No. 337]

177. (4)
Skeletal muscles can fatigue after prolonged use.
[New NCERT Class 11th Page No. 218]

178. (2)
Goitre is the enlargement of the thyroid gland, often caused by iodine deficiency.
[New NCERT Class 11th Page No. 334]

179. (1)

The synaptic cleft is the gap between two neurons where neurotransmitters are released to transmit impulses.

[New NCERT Class 11th Page No. 232]

180. (1)

The release of hormones from the posterior pituitary is under the direct neural regulation of the hypothalamus.

[New NCERT Class 11th Page No. 241]

181. (3)

The cranial meninges consist of three layers. The dura mater is the tough, outermost layer that lies just beneath the skull.

[New NCERT Class 11th Page No. 236]

182. (1)

Myelin sheath is produced by Schwann cell.

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183. (4)

- The anterior pituitary releases growth hormone (GH) in response to growth hormone-releasing hormone (GnRH).
- Pelvic girdles meet ventrally to form the pubic symphysis containing fibrous cartilage.
- The 8th, 9th, and 10th pairs of ribs do not directly attach to the sternum but connect to the seventh rib via hyaline cartilage.
- A single U-shaped bone called hyoid is present at the base of the buccal cavity.
- The number of mitochondria are few in white muscle fibers.

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184. (4)

- Low GH secretion causes pituitary dwarfism.
- Ribs have two articulation surfaces at the dorsal end, making them bicephalic.
- Pineal gland helps to regulate menstrual cycle.
- Pars intermedia secretes melanocyte stimulating hormone (MSH).

[New NCERT Class 11th Page No. 240]

185. (1)

Amoeboid movement	Leucocytes
Flagellar movement	<i>Euglena</i>
Myoglobin	Red muscle fibers
Thoracic vertebrae	12

[New NCERT Class 11th Page No. 218, 219, 227]

186. (4)

Unmyelinated nerve fibre is found in somatic neural system.

[New NCERT Class 11th Page No. 222]

187. (3)

The anterior pituitary produces GH, PRL, TSH, and ACTH hormones.

[New NCERT Class 11th Page No. 340]

188. (2)

Electrical synapses are rare in our system.

[New NCERT Class 11th Page No. 240]

189. (1)

The sliding filament theory explains skeletal muscle contraction, where actin slides over myosin, resulting in shortening of the muscle fibers.

[New NCERT Class 11th Page No. 221]

190. (1)

Osteoporosis is a age-related disorder characterised by decreased bone mass and increased chances of fractures.

[New NCERT Class 11th Page No. 227]

191. (3)

Ammonia is highly soluble in water and requires large quantities of water for its excretion to avoid toxicity, making it suitable for aquatic animals.

[New NCERT Class 11th Page No. 233]

192. (1)

Synaptic vesicles	Neurotransmitter
Cerebellum	Hindbrain
Corpora quadrigemina	Midbrain
Corpus callosum	Forebrain

[New NCERT Class 11th Page No. 236]

193. (4)

The coxal bone is made up of the ilium, pubis, and ischium.

[New NCERT Class 11th Page No. 221]

194. (4)
Neurons vary in structure (e.g., dendrites and axons) depending on their function.

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195. (2)
The sympathetic nervous system is part of the autonomic nervous system and prepares the body for "fight or flight" responses during stress.

[New NCERT Class 11th Page No. 231]

196. (1)
Cranial bones are 8 in number while facial bones are 14 in numbers.

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197. (1)

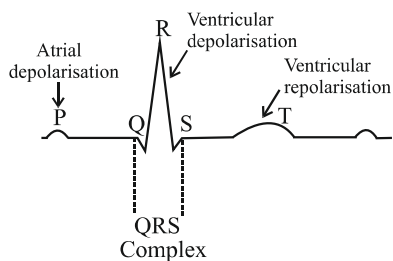


Fig: Diagrammatic representation of a standard ECG.

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198. (3)
Erythropoietin stimulates erythropoiesis, the production of red blood cells.

[New NCERT Class 11th Page No. 000]

199. (2)
Estradiol and progesterone are steroid hormones.

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200. (3)

- The sodium-potassium pump actively transports Na^+ out and K^+ into the neuron, maintaining the resting potential.
- The resting potential is primarily due to the high concentration of K^+ ions inside the neuron.

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